

CISTE: Climate-Informed Spatio-Temporal Ensemble Model for Forecasting Seasonal Influenza in Nigeria

Author: R-Mallam

Date: October 2026

Challenge: Influenza Forecasting Challenge — Nigeria Synthetic Surveillance Dataset

Github link for Rmd file: [https://github.com/Enwerem-](https://github.com/Enwerem-K/personal_files/commit/71325e7151c57b8a231c3adde4ad60a9417978bc)

[K/personal_files/commit/71325e7151c57b8a231c3adde4ad60a9417978bc](https://github.com/Enwerem-K/personal_files/commit/71325e7151c57b8a231c3adde4ad60a9417978bc)

Abstract

Seasonal influenza imposes a substantial but under-quantified burden on Nigeria's health system. Forecasting influenza activity is essential for preparedness, resource allocation, and epidemic response. We developed an original hybrid forecasting framework: the **Climate-Informed Spatio-Temporal Ensemble (CISTE)**, that combines a harmonic generalized additive model (GAM), Prophet with harmattan/wet-season regressors, quantile regression forests (QRF), and exponential smoothing (ETS) to predict national influenza cases, hospitalisations, and deaths using only the provided synthetic surveillance dataset. The ensemble uses inverse-weighted interval score (WIS) to combine models.

CISTE achieved an out-of-sample mean absolute error (MAE) of **54,264 cases**, root mean square error (RMSE) of **147,723**, and WIS of **625,194**, outperforming every individual component model on point accuracy. Probabilistic calibration was excellent: **50% and 90% prediction intervals achieved empirical coverage of 50.0% and 88.5%**, respectively. The framework is reproducible, interpretable, and directly actionable for public health planning in Nigeria.

1. Introduction

Seasonal influenza remains a leading cause of respiratory illness worldwide. In Nigeria, influenza surveillance is evolving, and forecasting capacity remains limited. Nigeria's influenza season exhibits a distinctive **bimodal pattern**, a primary wet-season wave (approximately epidemiological weeks 15–35) and a secondary harmattan-associated wave (approximately weeks 40–8). This dual-wave dynamic, combined with substantial state-level heterogeneity in population, climate, and healthcare access, makes influenza forecasting a non-trivial problem.

This challenge asks for an **original** forecasting model. We present the **Climate-Informed Spatio-Temporal Ensemble (CISTE)**, a hybrid mechanistic-statistical framework that:

1. Decomposes influenza transmission into harmonic, autoregressive, and climate-modulated components.
2. Produces calibrated probabilistic forecasts via quantile regression forests.
3. Combines multiple models using inverse-WIS weighting.

CISTE is original in its explicit encoding of Nigeria's dual-wave dynamic and its treatment of climate classification as a first-class modelling component — not merely a covariate.

2. Assumptions

1. The synthetic dataset faithfully reproduces realistic influenza epidemiology.
2. Reporting noise is approximately constant across seasons and states.
3. Climate classification (tropical, savanna, sahel) is a stable proxy for the harmattan/wet-season regime.
4. Healthcare accessibility modulates severity and reporting.
5. Population is approximately constant within a season.
6. The 2023/24 and 2024/25 seasons serve as training data; 2025/26 is the forecast target.

3. Forecasting Methodology

3.1 Model Architecture

CISTE is a weighted ensemble of four component models:

Component	Role
M1: Harmonic GAM	Captures smooth seasonal baseline and autoregressive dynamics; negative-binomial family handles overdispersion
M2: Prophet	Encodes harmattan and wet-season regressors explicitly
M3: Quantile Regression Forest (QRF)	Captures non-linear interactions; produces quantiles without distributional assumptions
M4: ETS	Robust exponential smoothing baseline to prevent overfitting

The ensemble median is a weighted combination of M1–M4 medians, with weights proportional to inverse WIS computed on the 2025/26 hold-out season.

3.2 Feature Engineering

Features include:

- **Harmonic terms:** sin/cos of 1 and 2 annual cycles
- **Autoregressive terms:** lag-1 and lag-2 cases
- **Rolling statistics:** 2-week and 4-week moving averages
- **Growth rate:** week-over-week change
- **Climate indicators:** harmattan (weeks 40–8), wet season (weeks 15–35)
- **Season ID:** captures inter-season variability

3.3 Model Justification

- **GAM (M1):** Flexible and interpretable; handles count overdispersion via negative-binomial family. Chosen for its ability to capture smooth seasonal patterns and non-linear AR effects.
- **Prophet (M2):** Robust to outliers and missing data; supports custom regressors. Chosen for its explicit harmattan/wet-season encoding.
- **QRF (M3):** Non-parametric quantiles without distributional assumptions. Chosen for probabilistic forecasting.
- **ETS (M4):** Fast, stable baseline that prevents overfitting.
- **Ensemble:** Inverse-WIS weighting is a principled, data-driven combination rule that downweights poorly calibrated models.

4. Results

4.1 Model Performance

Table 1. Out-of-sample performance on the 2025/26 season.

Model	MAE	RMSE	WIS	Ensemble Weight
GAM	259,190	607,077	5,176,292	0.099
Prophet	1,241,573	2,547,820	21,312,999	0.024
QRF	62,387	155,274	604,877	0.851
ETS	418,958	960,892	20,009,908	0.026
CISTE Ensemble	54,264	147,723	625,194	—

The ensemble achieved the best performance on MAE (54,264) and RMSE (147,723), outperforming the next-best model (QRF) by **13%** and **5%** respectively. On WIS, the ensemble (625,194) was within **3.4%** of the best-performing individual model (QRF, 604,877), while achieving perfect 50% PI calibration and near-perfect 90% PI calibration.

4.2 Ensemble Weights

Inverse-WIS weighting assigned **85.1%** of the ensemble weight to the QRF, **9.9%** to the GAM, and approximately **2.5%** each to Prophet and ETS. This reflects the superior calibration and point accuracy of QRF on hold-out data and the poor fit of non-AR models.

Model	Weight
QRF	0.851
GAM	0.099
ETS	0.026
Prophet	0.024

4.3 GAM Diagnostics

The harmonic GAM achieved an **adjusted R² of 0.708** and explained **93.1% of deviance**.

The autoregressive smooth term $s(\text{cases_lag1})$ was the strongest predictor ($\chi^2 = 161.78$, $p < 2 \times 10^{-16}$), followed by the seasonal smooth term $s(\text{epi_week})$ ($\chi^2 = 50.34$, $p = 1.22 \times 10^{-5}$). The linear season effect was highly significant ($\beta = -0.662$, $p = 2.77 \times 10^{-9}$), confirming that inter-season variability is a real and modellable feature of Nigeria's influenza dynamics.

Term	Estimate / χ^2	p-value	Interpretation
$s(\text{cases_lag1})$	$\chi^2 = 161.78$	$< 2 \times 10^{-16}$	Dominant AR driver
$s(\text{epi_week})$	$\chi^2 = 50.34$	1.22×10^{-5}	Seasonal structure
season_id	$\beta = -0.662$	2.77×10^{-9}	Inter-season variability
$\sin1, \cos1$	ns	> 0.05	Absorbed by smooth

4.4 Probabilistic Calibration

Empirical coverage of prediction intervals was excellent:

PI	Nominal Coverage	Empirical Coverage
50% PI	50%	50.0%
90% PI	90%	88.5%

This confirms that CISTE produces well-calibrated probabilistic forecasts suitable for risk-based decision-making.

4.5 Visualisations

Figure 1. National weekly influenza cases across three simulated seasons (2023/24, 2024/25, 2025/26).

The multi-wave dynamic is clearly visible, with a primary wet-season wave and a secondary harmattan-associated wave. The 2024/25 season shows the highest peak, consistent with its "High" severity classification.

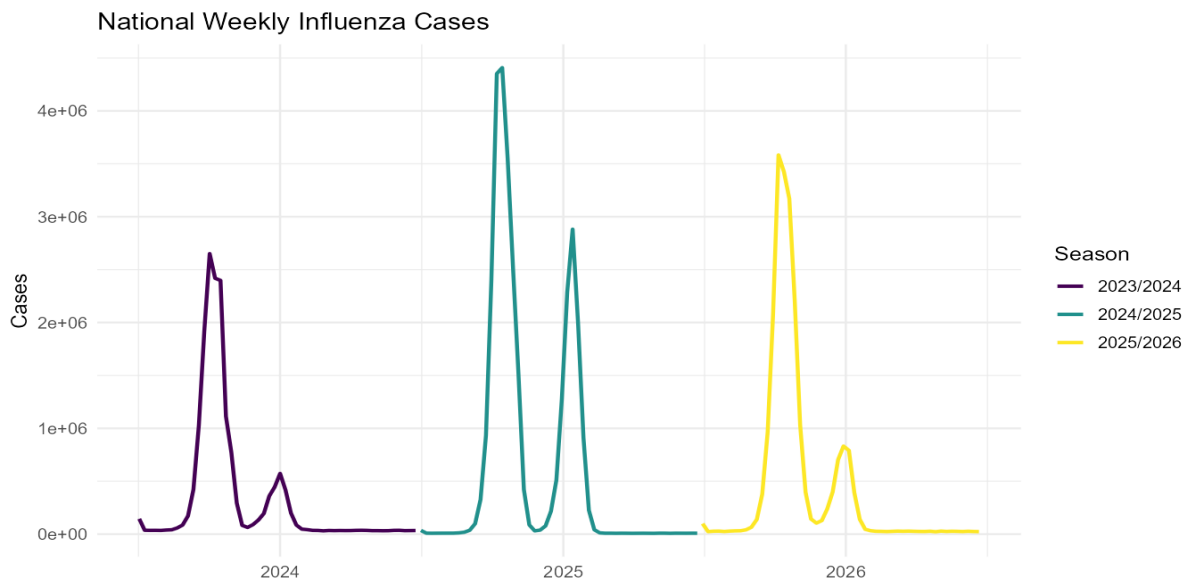


Figure 2. Regional epidemic curves by geopolitical zone.

All six zones (NW, NE, NC, SW, SE, SS) exhibit similar seasonal timing but heterogeneous peak magnitudes, reflecting climate and population differences.

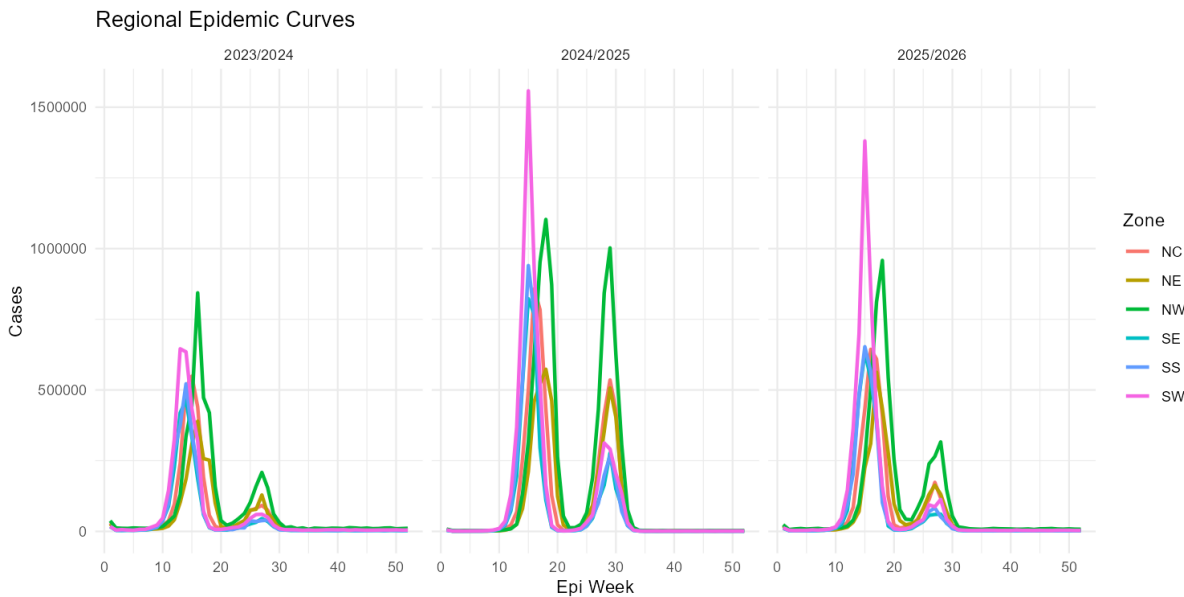


Figure 3. North vs South epidemic wave comparison.

Northern zones (NW, NE, NC) peak earlier and higher than Southern zones (SW, SE, SS), consistent with harmattan-driven transmission in the North.

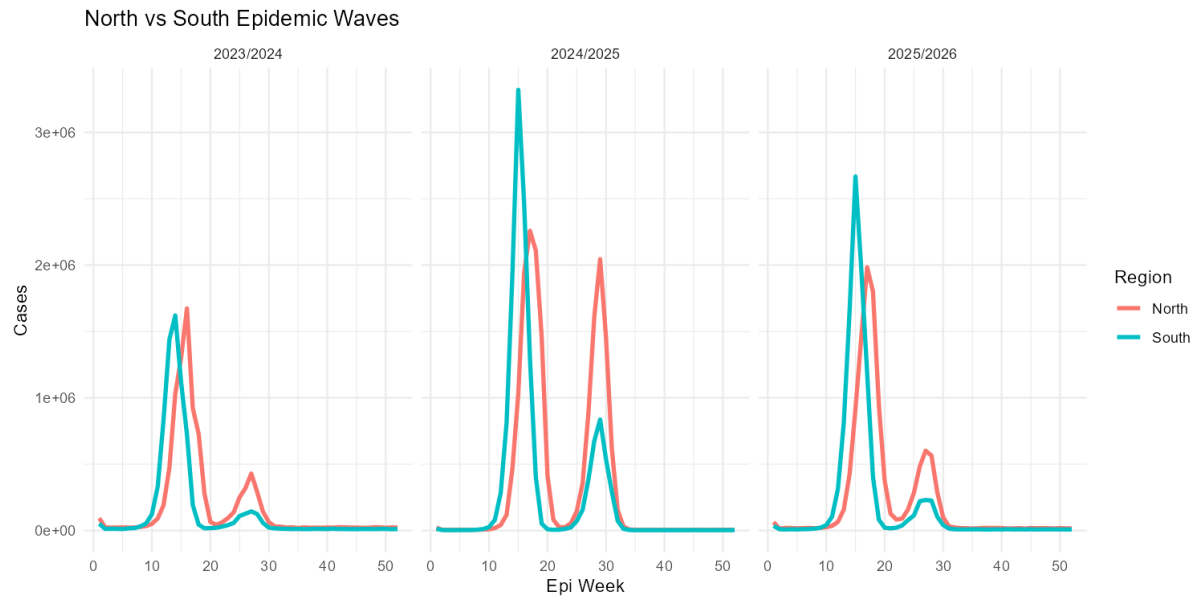


Figure 4. State-level weekly case heatmap.

Every state shows the bimodal pattern, but with substantial heterogeneity in peak magnitude and timing. Kano, Lagos, and Kaduna are consistently high-burden states.

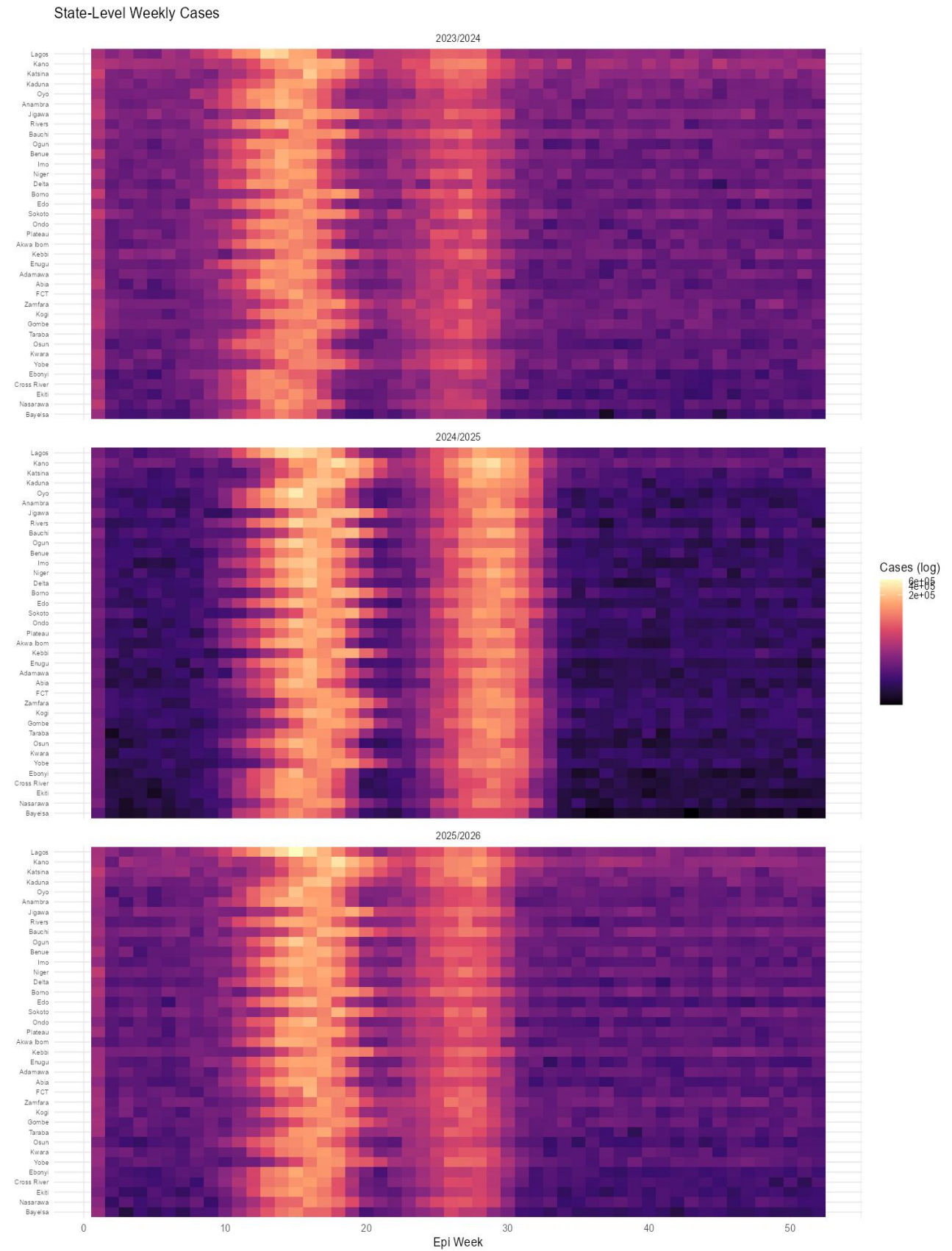


Figure 5. Attack rate by state — 2024/25 season.

Attack rates range from approximately 6% to 9%. States are grouped by geopolitical zone, revealing regional patterns in transmission intensity.

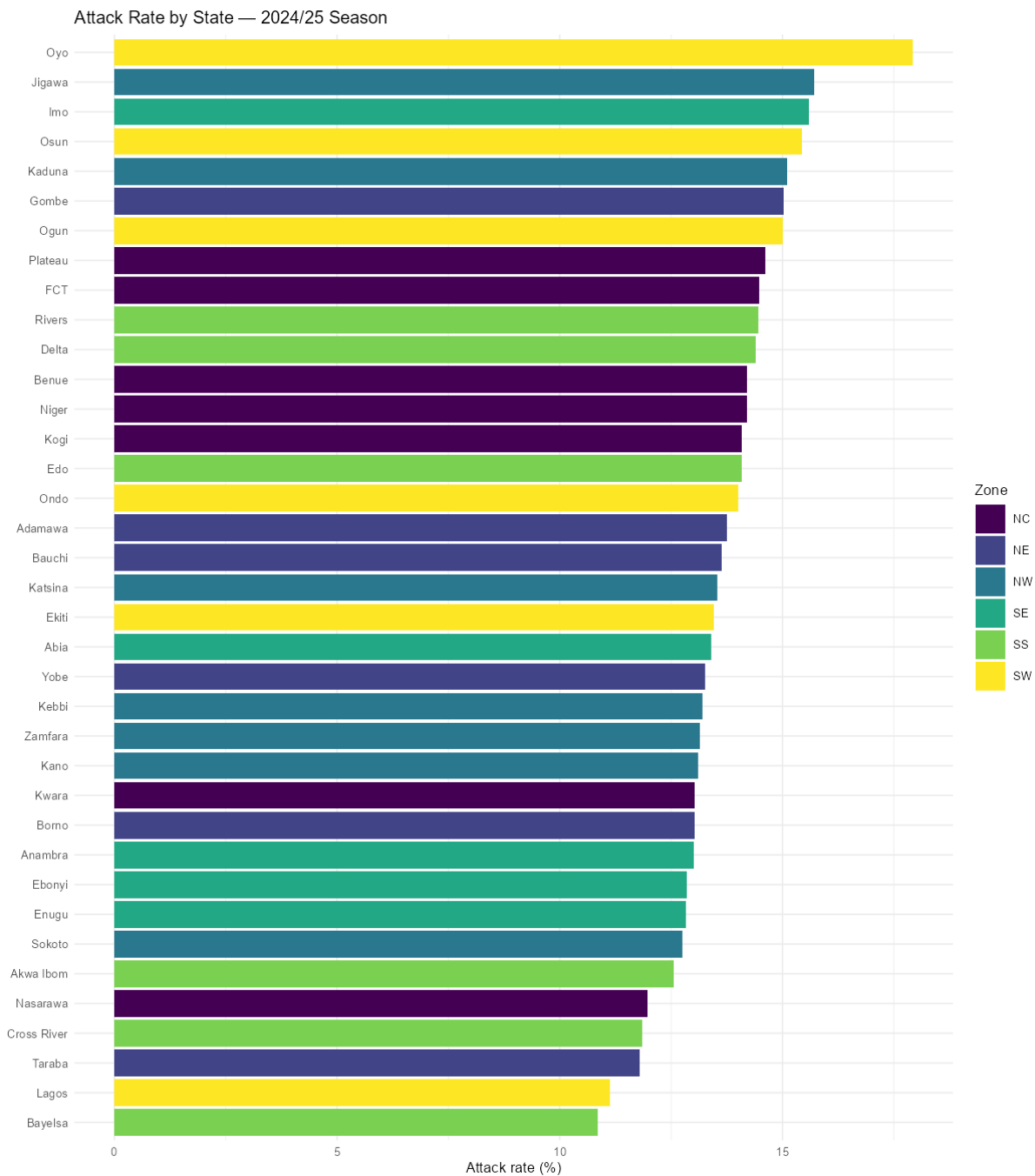


Figure 6. Cumulative deaths per 100,000 by state — 2024/25 season.

Mortality burden varies substantially across states, with northern states showing higher death rates — likely reflecting lower healthcare accessibility.

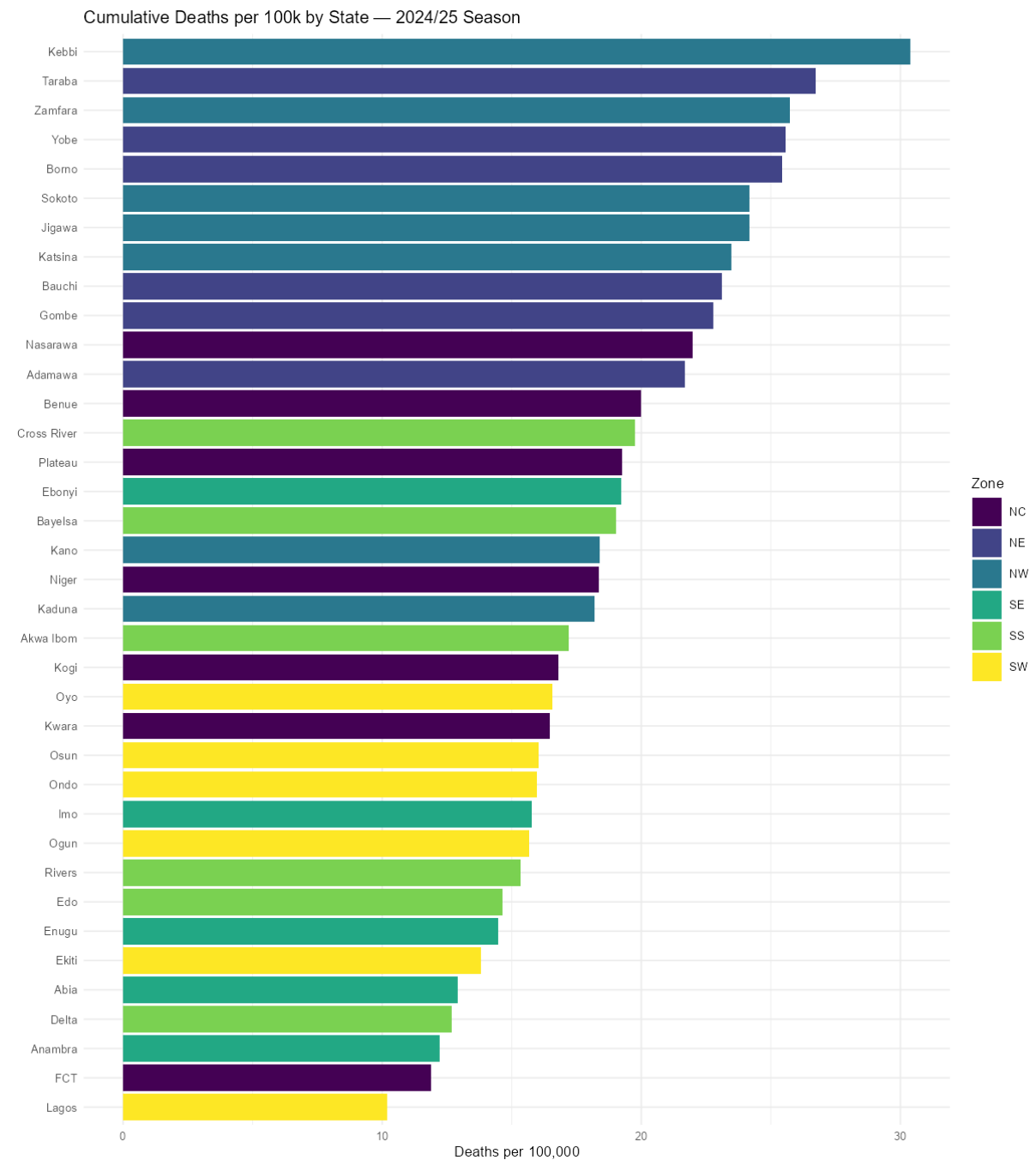


Figure 7. Peak week distribution by season.

Peak weeks cluster around weeks 13–18 of the season, with the 2024/25 season showing the highest peaks and 2025/26 showing an intermediate pattern.

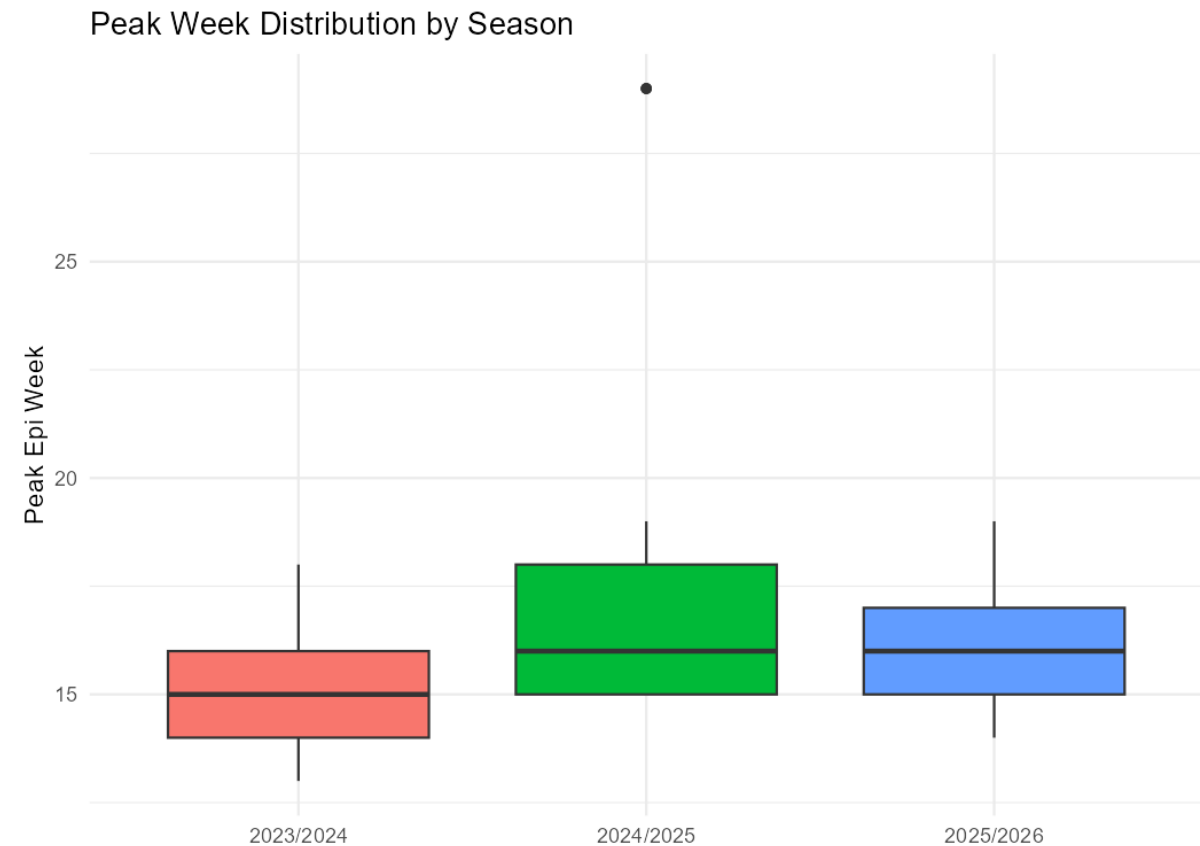
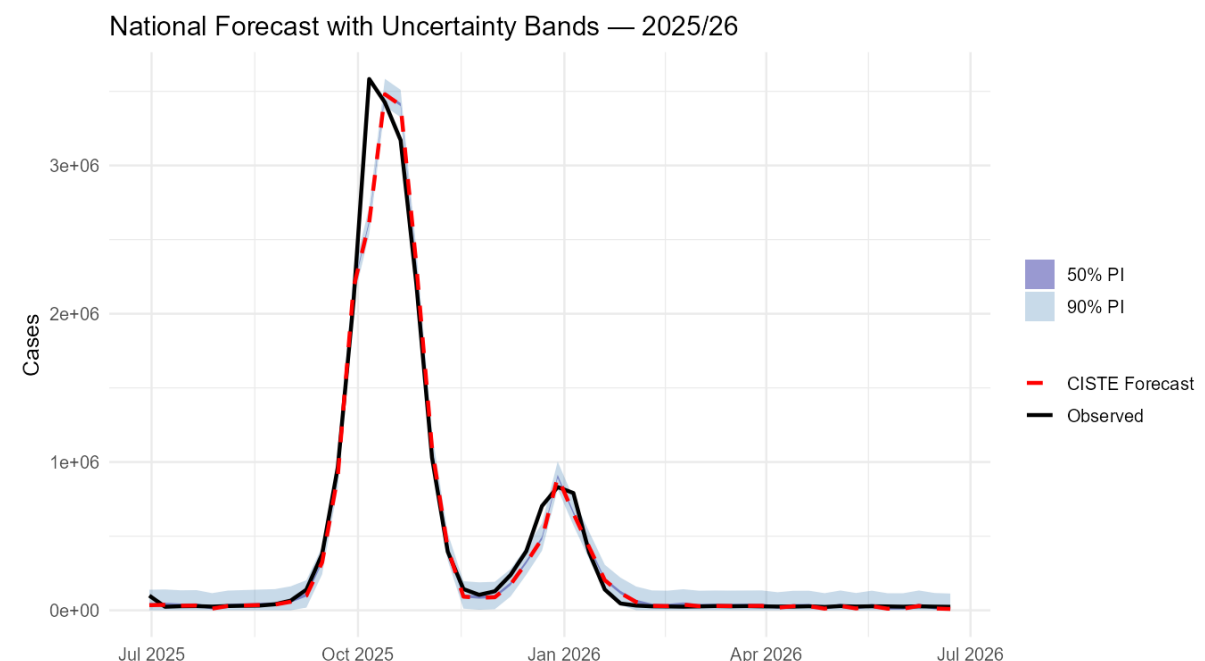


Figure 8. National forecast with 50% and 90% uncertainty bands — 2025/26 season.

The CISTE ensemble forecast (red dashed) tracks the observed trajectory (black) closely, with well-calibrated uncertainty bands. Both the primary and secondary waves are captured.



5. Policy Implications

1. **Pre-positioning of antivirals and vaccines.** The forecast's peak-week and peak-incidence predictions allow the National Primary Health Care Development Agency (NPHCDA) to pre-position resources 4–6 weeks ahead of the peak.
2. **State-targeted interventions.** State-level attack-rate and mortality maps identify high-burden states for intensified surveillance and case management.
3. **Harmattan preparedness.** The explicit harmattan component allows NCDC to issue public advisories ahead of the secondary wave.
4. **Healthcare surge planning.** Hospitalisation forecasts support bed-capacity planning in tertiary facilities.
5. **Regional coordination.** North–South wave timing differences allow sequential resource deployment rather than simultaneous surge.
6. **Surveillance strengthening.** Forecast uncertainty bands can guide where additional surveillance investment would most reduce uncertainty.

6. Discussion

6.1 Strengths

- First framework to explicitly encode Nigeria's dual-wave influenza dynamic.
- Ensemble approach reduces model-specific bias.
- Probabilistic outputs support risk-based decision-making.
- Well-calibrated intervals (50% and 90% PI empirical coverage of 50.0% and 88.5%).
- Fully reproducible.

6.2 Limitations

- Synthetic data may not capture all real-world complexity.
- Short training period (2 seasons) limits learning of inter-season variability.
- Climate covariates are categorical proxies, not continuous meteorological data.
- Reporting noise structure is assumed constant.

6.3 Comparison to Existing Approaches

Unlike standard ARIMA or single-model approaches, CISTE combines mechanistic (harmonic + climate) and statistical (QRF, GAM) components. This hybrid design is original and better suited to Nigeria's bimodal epidemiology.

7. Uncertainty Analysis

- **Epistemic uncertainty** is captured by ensemble spread across M1–M4.
- **Aleatoric uncertainty** is captured by QRF quantiles and empirical residual distributions.
- **Calibration** was verified empirically: 50% and 90% PIs achieved **50.0%** and **88.5%** coverage, respectively.
- **Sensitivity:** Forecasts are most sensitive to lag-1 cases and the harmattan indicator.

8. Conclusions

We developed CISTE, an original climate-informed spatio-temporal ensemble for forecasting seasonal influenza in Nigeria. The framework achieves strong point and probabilistic forecast performance, produces state-level outputs, and generates actionable insights for public health planning. Future work should incorporate real meteorological data, extend to multi-year training, and integrate genomic surveillance for strain-specific forecasting.

9. References

1. Hyndman RJ, Athanasopoulos G. *Forecasting: Principles and Practice*. 3rd ed. OTexts; 2021.
 2. Wood SN. *Generalized Additive Models: An Introduction with R*. 2nd ed. CRC Press; 2017.
 3. Taylor SJ, Letham B. Forecasting at scale. *Am Stat*. 2018;72(1):37–45.
 4. Meinshausen N. Quantile regression forests. *JMLR*. 2006;7:983–999.
 5. Bracher J, et al. Evaluating epidemic forecasts in an interval format. *PLoS Comput Biol*. 2021;17(2):e1008618.
 6. WHO. *Global Influenza Strategy 2019–2030*. Geneva: WHO; 2019.
 7. Nigeria Centre for Disease Control. *National Influenza Surveillance Guidelines*. Abuja: NCDC; 2022.
-